TANYA KUMARI

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EDUCATION

Silicon Institute of Technology, Sambalpur **Expected 2027**
B.Tech — Computer Science and Engineering
Manav Bharti National School
AISSCE (Science Stream)

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL
Frontend: React.js, HTML5, CSS3, Responsive Design, Hooks, Context API, Lazy Loading, Memoization
Backend: Node.js, Express.js, FastAPI, Spring Boot, REST APIs, JWT Auth, Kafka, WebSockets (Socket.io), Redis
Databases: MongoDB, Mongoose ODM, H2, Schema Design, Indexing, Aggregation Pipelines
AI / ML / Data: Scikit-learn, XGBoost, Pandas, NumPy, ROC-AUC, Generative AI, LLMs, Big Data, Hadoop, Docker
DevOps & Tools: Git, GitHub, AWS, Vercel, Render, Jupyter, Google Colab, VS Code
Concepts: DSA, OOP, DBMS, System Design, Cybersecurity, Message Queuing, Distributed Computing, Agile/Scrum

PROJECTS

ManSaathi — RAG-Powered, Model-Agnostic Therapeutic AI Chatbot **2026**
- Built a Retrieval-Augmented Generation (RAG) chatbot delivering empathetic, context-grounded therapeutic conversations for accessible mental health support.
- Engineered a model-agnostic architecture decoupling chatbot logic from any single LLM, enabling plug-and-play integration across multiple LLM backends.
- Implemented a retrieval pipeline with vector-based knowledge base indexing to ground responses in curated therapeutic content and reduce hallucinations.
- Developed at the Frugal Innovation Joylab, CRDT, IIT Delhi, during a 2-month offline AI/ML research internship under Prof. Vijayaraghavan M Chariar.

Smart API Rate Limiter & Discovery System — Node.js, Express **2026**
- Built a real-time API traffic monitoring platform with sliding-window rate limiting and Redis-based distributed token bucket across instances.
- Engineered configurable plug-and-play Express middleware for seamless rate-limiting integration into existing microservice-based architectures.
- Designed automated API discovery modules with dynamic route scanning, metadata extraction, and centralized documentation generation.
- Implemented structured request logging to trace dropped requests and flag potential DDoS patterns under high concurrency stress.

Customer Churn Prediction — ML Pipeline & FastAPI Deployment **2024**
- Built an end-to-end ML pipeline on ~10,000 Telco records achieving 89% accuracy with oversampling for class imbalance correction.
- Analyzed ROC-AUC scores, precision-recall curves, and confusion matrices to quantify model success and minimize false negatives.
- Containerized the complete ML pipeline and FastAPI backend using Docker for environment consistency and reproducible deployments.
- Designed an interactive web dashboard for non-technical stakeholders to input parameters and receive churn probabilities.

INTERNSHIPS & PROFESSIONAL TRAINING

AI/ML Research Intern — Centre for Rural Development & Technology (CRDT), IIT Delhi **May – Jul 2026**
- Completed an offline AI/ML internship under Prof. Vijayaraghavan M Chariar in the Frugal Innovation Joylab, CRDT, IIT Delhi.
- Built “ManSaathi,” a RAG-Powered, Model-Agnostic Therapeutic AI Chatbot designed to deliver accessible, empathetic mental health support.
- Implemented Retrieval-Augmented Generation (RAG) pipelines to ground chatbot responses in curated therapeutic knowledge bases.
- Designed a model-agnostic architecture enabling seamless integration with multiple LLM backends for flexible, scalable deployment.

Vinternship — AKSIANS Program, IIT Ropar (NPTEL) **2026**
- Engineered full-stack web applications using the complete MERN stack to solve complex, real-world industry case studies effectively.
- Developed responsive, dynamic, and accessible user interfaces with React, significantly improving UX and cross-device compatibility.
- Designed and implemented secure, well-documented RESTful APIs using Node.js and Express for backend routing and DB management.
- Managed database architecture, schema design, and data flow using MongoDB, ensuring highly scalable and maintainable data persistence.

AI Fundamentals — Agriculture Dataset, Vicharanshala.ai **2026**
- Evaluated and compared ML model performance using Accuracy, Precision, Recall, and F1-score to select the most optimal algorithms.
- Conducted comprehensive Exploratory Data Analysis on complex agricultural datasets to uncover hidden feature correlations and trends.
- Optimized machine learning algorithm hyperparameters through systematic tuning to improve agricultural forecasting model accuracy.
- Visualized model predictions against actual agricultural outcomes using detailed reporting to assess real-world viability and impact.

CERTIFICATIONS

- **IIT Delhi (CRDT)** — Offline AI/ML Internship, Frugal Innovation Joylab (May–Jul 2026)
- **IBM (Coursera)** — Generative AI: Boost Your Cybersecurity Career (May 2025)
- **UC San Diego (Coursera)** — Introduction to Big Data (Dec 2024, Grade: 96.40%)
- **U of Michigan (Coursera)** — Intro to Data Science in Python (Dec 2024, Grade: 94.80%)
- **NPTEL (Elite)** — Artificial Intelligence: Concepts and Techniques (Jul–Oct 2025, 6,785 certified)
- **NPTEL (Elite)** — Python for Data Science (Jan–Feb 2025, Score: 70%, Assignments: 25/25)
- **IIT Ropar** — AKSIANS Vinternship Program (NPTEL)
- **Vicharanshala.ai** — AI Fundamentals (Agriculture Dataset)